1 The Fundamental Theorem of Calculus

Definition 1.1 (The integral)**.** The *integral* is a mapping $\int : \mathscr{R}[a,b] \to \mathbb{R}$ which assigns a function $f : [a,b] \to \mathbb{R}$ to a real number $\int_a^b f$ in such a way that satisfies the following axioms:

(I0) $\mathscr{C}[a,b] \subseteq \mathscr{R}[a,b] \subseteq \mathscr{B}[a,b]$.
(I1) If $f = c$ is a constant function, then $f \in \mathscr{R}[a,b]$ and $\int_a^b c = c(b-a)$.
(I2) If $f_1, f_2 \in \mathscr{R}[a,b]$ and $f_1(x) \leq f_2(x)$ for all $x \in [a,b]$, then $\int_a^b f_1 \leq \int_a^b f_2$.
(I3) If $a \leq c \leq b$, then $f \in \mathscr{R}[a,b]$ if and only if $f \in \mathscr{R}[a,c]$ and $f \in \mathscr{R}[c,b]$, in which case $\int_a^b f = \int_a^c f + \int_c^b f$.

If $a > b$, then we define $\int_a^b f = -\int_b^a f$. It then follows that for *any* a, b, and c, $\int_a^b f = \int_a^c f + \int_c^b f$.

Theorem 1.2 (Fundamental Theorem of Calculus)**.** *Let $f \in \mathscr{R}[a,b]$ be any integrable function. For $x \in [a,b]$ define $\mathscr{F} : [a,b] \to \mathbb{R}$ by $\mathscr{F}(x) = \int_a^x f$. Then:*

(a) The function $\mathscr{F}$ is continuous at every $c \in [a,b]$.

(b) If f is continuous at $c \in [a,b]$, then $\mathscr{F}$ is differentiable at c, and $\mathscr{F}'(c) = f(c)$.

(c) If f is continuous on $[a,b]$ and F is any antiderivative of f (i.e., a function $F : [a,b] \to \mathbb{R}$ such that $F'(x) = f(x)$ for all $x \in [a,b]$), then $\int_a^b f = F(b) - F(a)$.

1.1 Proof of part (a)

To prove that $\mathscr{F}$ is continuous at $c \in [a,b]$, we need to show that, given $\varepsilon > 0$, there exists $\delta > 0$ such that $|x-c| < \delta$ implies $|\mathscr{F}(x) - \mathscr{F}(c)| < \varepsilon$. By (I0), there exists $M \geq 0$ such that $|f(x)| \leq M$ for all $x \in [a,b]$.

Let $\varepsilon > 0$.

1. Show that $\mathscr{F}$ is continuous if $M = 0$.

2. Show that $\mathscr{F}(x) - \mathscr{F}(c) = \int_c^x f$.

3. Show that $\mathscr{F}$ is continuous if $M > 0$. (*Hint*: You will need to consider the cases $x > c$ and $x < c$ separately.)

1.2 Proof of part (b)

Suppose f is continuous at $c \in [a,b]$. We need to show that

$$\begin{aligned}\mathscr{F}'(x) &= \lim_{x\to c} \frac{\mathscr{F}(x) - \mathscr{F}(c)}{x-c} \\ &= \lim_{x\to c} \frac{\int_c^x f}{x-c} \\ &= f(c)\end{aligned}$$

exists for each $c \in [a,b]$ (with the limit above being replaced by appropriate one-sided limits when $x = a$ or $x = b$). Equivalently, we can show that $\lim_{x\to c}(\frac{\int_c^x f}{x-c} - f(c)) = 0$.

3. Given $\varepsilon > 0$, show that there exists $\delta > 0$ such that $|x-c| < \delta$ implies $f(c) - \varepsilon < f(x) < f(c) + \varepsilon$.

4. Show that, if $x > c$, then $|x-c| < \delta$ implies

$$f(c) - \varepsilon < \frac{\int_c^x f(x)}{x-c} < f(c) + \varepsilon$$

and therefore $\left| \frac{\int_c^x f(x)}{x-c} - f(c) \right| < \varepsilon$.

5. Show that, if $x < c$, then $|x-c| < \delta$ implies the same inequality, which completes the proof.

1.3 Proof of part (c)

6. Prove part (c). Note that part (b) says that, if f is continuous, then $\mathscr{F}$ is an antiderivative of f. Any other antiderivative F for f must therefore be of the form $F(x) = \mathscr{F}(x) + c$ for some constant c.